An Evolutionary Approach to Traffic Assignment

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***Abstract*—Traffic assignment is an important stage in traffic modeling. Most of the existing approaches are based on finding an approximate solution to the user equilibrium or to the system optimum, which can be computationally expensive. In this paper we use a genetic algorithm to compute an approximate solution (routes for the trips) that seeks to minimize the average travel time. To illustrate this approach, a non-trivial network is used, departing from binary route choice scenarios. Our result shows that the proposed approach is able to find low travel times, without the need of recomputing shortest paths iteratively.**

I. Introduction

Traffic assignment is an important stage in the task of modeling a transportation system. It connects the physical infrastructure and the demand that is going to use it, i.e., it assign trips to each link of the road network. This is done by assuming that each user will perform adaptive route choices until the so-called user equilibrium is found (Wardrop's first principle), or by computing the assignment that corresponds to the system optimum (Wardrop's second principle). This states that under social equilibrium conditions, traffic should be arranged in congested networks in such a way that average (or total) travel cost is minimized.

One issue with traffic assignment is that even if each driver has a small set of routes to choose to go from A to B, the number of combinations of these routes (one for each driver) can be very high. The problem is not only complex due to this fact, but also because each trip affects the cost of many others since they use restricted resources (the links themselves). Section II discusses some methods that have been employed to solve this problem in an approximate way.

In this paper we propose the use of a GA-based method that, although not guaranteed to find the optimum solution (from the system's point of view) for the trip assignment, has the advantage of being simple to implement and finds good solutions within minutes, without the need of recomputing shortest paths iteratively (as some of the approximate solutions do).

Moreover, for sake of illustration, the approach is tested in a non-trivial scenario, in which some links are shared by many trips, thus producing congestion if too many trips use these links. Finally, the scenario we tackle departs from a single flow and two or three choices, as, e.g. [1], [2].

Apart from background concepts, methods, and related work on trip assignment, which are discussed in the next two sections, Section IV describes the proposed approach and the scenario used to illustrate it. Results are shown and analyzed in Section V, while Section VI presents the concluding remarks and points to future research.

II. Traffic Assignment Methods

In this section, basic concepts about traffic (or trip) assignment methods are given. For an extensive explanation, please refer to Chapter 10 in [3] or to Chapter 4 in [4].

A road network can be represented as a graph $G = (V, E)$, where V is the set of vertices that represent the intersections of the network, and E is a set of directed arcs, describing the existing road segments as directed connections between pairs of vertices. Each link $l_k \in L$ has a cost c_k, which is given by a function that takes as input attributes such as length, toll, free-flow speed (and hence, free-flow travel time), capacity, current volume, etc.

A route r_p is defined by a sequence of connected vertices $(v_0, v_1, v_2, ...)$. The cost of each r_p is the sum of the costs of all links l_k that connect these vertices.

We remark that trip assignment is generally done in a macroscopic way, i.e., contrarily to a microscopic traffic simulator where the actual movement of vehicles is considered, in a macroscopic simulation, vehicles are considered in an abstract way. One well know abstraction is the concept of volume-delay functions (VDFs) or cost-flow relationship. These functions abstract physical measures such as gap between vehicles etc. in a relationship that accounts for congestion effects, i.e., how over-capacity of a given volume or flow in a link affects the speed and travel times (costs of delays).

VDFs receive as input the flows in the whole network, i.e., they consider the interactions between flows that use the network at the same time, and output the corresponding delays that may occur. As an example of a VDF, one can consider the following: $t_k = t_{k_0} + 0.02 \cdot q_k$. Here, t_k is the travel time on link k, t_{k_0} is the travel time per unit of time under free flow conditions, and q_k is the flow using link k. This means that the travel time in each link increases by 0.02 of a minute for each vehicle/hour of flow. Other similar VDFs exist (see [3] for instance).

Given a demand T_{ij} for trips between origin i and destination j, there are several schemes to assign these trips to the links of a road network. For example, given VDFs for each link in the network, a goal of these approaches is to approximate the equilibrium conditions as stated by Wardrop [5]: "under equilibrium conditions traffic arranges itself in

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congested networks such that all used routes between an OD pair have equal and minimum costs while all those routes that were not used have greater or equal costs". This is Wardrop's first principle, also known as Wardrop's equilibrium or user equilibrium.

Thus, given a traffic network, the assignment from the point of view of the user equilibrium can be analytically stated as an optimization problem: find all flows from each OD pair s.t. only paths with minimal costs have a nonzero flow assigned to them, which corresponds to Wardrop's first principle. For a mathematical formulation of this problem, the reader is referred to Chapter 2 in [6], as well as to [7].

One problem with this scheme is that it is not possible to solve the equilibrium flows algebraically, except for very simple cases (e.g., two or three links connecting a single OD pair). Thus, approximate solutions to the user equilibrium were proposed.

Such approximate solutions are discussed later in this section. Before, it is important to introduce a general procedure that underlies any of the assignment schemes. Indeed, each assignment scheme discussed before has several steps that must be treated in turn: (i) to identify a set of routes that might be considered attractive to drivers; (ii) to assign suitable proportions of the trip matrix to these routes; this results in flows on the links of the network; (iii) to search for convergence: many techniques follow an iterative pattern of successive approximations to an ideal solution (e.g., the user equilibrium).

The first step can be accomplished with any variant of the Dijkstra algorithm for shortest paths. This step is also known as tree-building step. However, normally these paths are generated based on a first-approximation or an estimated cost function (e.g., one that considers no congestion, i.e., only free-flow travel times are considered) because the real cost is not known, given that it depends on the route choices of all users. Therefore, under congestion, the second aforementioned step must be performed iteratively, until some sort of convergence is reached.

Next, some classical trip assignment approaches are discussed.

The typical approach to trip assignment under no congestion is to assign all trips to the route with minimum cost, on the basis that these are the routes travelers would rationally select. If T_{ij} is the given demand between origin i and destination j, for the minimum cost route $r^{\star}$, $T_{ijr^{\star}} = T_{ij}$, whereas $T_{ijr} = 0$ for all other routes. This procedure is known as "all-or-nothing" assignment: this scheme assigns all trips between nodes i and j to the same links (because, as mentioned, this scheme assumes no congestion).

For route assignment under congestion (i.e., the capacity of a link k can be surpassed and, as such, a VDF is necessary to account for the effects of the over-capacity), mainly two iterative methods can be used. The first is to load the network incrementally in n stages, e.g., assigning a given fraction p_n (e.g., 10%, 20%, etc.) of the total demand (for each OD pair) at each stage. Further fractions are then assigned based on the newly computed link costs. This procedure continues until 100% of the demand is assigned. Typical values for fractions p_n are 0.4, 0.3, 0.2, and 0.1.

An algorithm for this is the following (adapted from [3]):

1) select an initial set of current link costs (usually the free-flow travel times); initialize flows at all links k: $V_k = 0$; select a fraction p_n of the trip matrix T such that $\sum_n p_n = 1$; make $n = 0$.
2) build the set of minimum cost trees (one for each origin) using the current costs; $n \leftarrow n + 1$.
3) load $T_n = p_n T$ all-or-nothing trips to these trees, obtaining a set of auxiliary flows F_k; accumulate flows on each link: $V_k^n = V_k^{n-1} + F_k$.
4) calculate a new set of current link costs based on flows V_k^n; if not all fractions of T have been assigned, proceed to step 2.

It must be remarked that there is no guarantee that this algorithm converges to the user equilibrium, no matter how small each p_n is. This procedure has the drawback that once a flow has been assigned to a link, due to the accumulative nature (see step 3), it is never removed. Thus, in case an arbitrarily low over-capacity is assigned to a link, then it prevents the convergence to the optimum solution. However, it is very easy to implement.

The other approach is to start from some initial values for the link costs and find the minimum cost routes. Trips are then assigned to these routes. New costs are computed and this cycle is repeated until there is no significant change in link or route volumes. For instance, in the method of successive averages, the flow at the n-th iteration is calculated as a linear combination of the flow on the previous iteration and an auxiliary flow resulting from an all-or-nothing assignment in the n-th iteration.

This can be formalized as the following procedure (again, adapted from [3]):

1) select an initial set of current link costs (usually the free-flow travel times); initialize flows at all links k: $V_k = 0$; make $n = 0$.
2) build the set of minimum cost trees (one for each origin) using the current costs; $n \leftarrow n + 1$.
3) load the whole of the matrix T all-or-nothing to these trees obtaining a set of auxiliary flows F_k.
4) calculate the current flows as: $V_k^n \leftarrow (1-\phi)V_k^{n-1} + \phi F_k$, with $0 \leq \phi \leq 1$.
5) calculate a new set of current link costs based on V_k^n; if no V_k^n has changed significantly in two consecutive iterations, stop; otherwise proceed to step 2 (or, alternatively, use a maximum number of iteration).

The last step of the method admits several ways to fix the value of ϕ. A useful one is to make $\phi = 1/n$. There is a proof that this produces solutions convergent to the user equilibrium but this may take many iterations, thus being very inefficient.

Note that both iterative methods to approximate the user equilibrium are based on the all-or-nothing scheme. Thus, a certain number of trips is assigned to the same links.

III. Related Work

A number of works from transportation planning and economics, as well as from mathematics and operations research, physics, and computer science deal with this problem. Computer science plays a role when it comes to solving large-scale road network problems.

These approaches can be roughly divided in two classes: assignment performed from a global point of view, where a central authority is in charge of doing the assignment of routes to trips; and route choice from the individual (or agent) point of view. The GA-based approach proposed here lies in the former class. Next, we focus on the most relevant approaches in each class, starting with the former.

In [8], it is shown that trip assignment schemes that are based on steady-state flow conditions of the road network are adequate only for analysis of long-term strategic planning horizons, but not for tactical measures that are of interest in applications around intelligent transportation systems. However, the author also recognizes the challenges of finding an analytic representation that satisfies the laws of physics and traffic sciences, while also being mathematically tractable. Therefore they propose a simulation-based approach for the dynamical traffic assignment (DTA) problem. In DTA one goal is to describe how flows develop not only spatially but also temporally in the network. This means that DTA considers road users that depart from an origin to a destination at different times. Hence, they experience different travel times and, as such, the user equilibrium condition applies only to travelers who are assumed to depart at the same time between the same OD pair.

DTA is also the focus of [9] in which the authors propose a predictive DTA model, also based on simulation and combined with the method of successive averages.

Henn [10] proposes a fuzzy-based method to take the imprecisions and the uncertainties of the road users into account. These predict costs for each path based on a fuzzy subset that can represent imprecision on network knowledge, as well as uncertainty on traffic conditions.

Assignment is combined with signal optimization in [11] but the focus of their GA is not on the trips.

GA applied in a microscopic simulation based approach is discussed by us in [12], for the same network we use in the present paper. We remark that in the microscopic approach, the actual (physical) movement of cars is simulated. Thus, run times tend to be much longer than when a macroscopic approach is used (as we do in the present paper).

As for the aforementioned second class (individual route choice), a natural way to represent the problem is to model it using an agent-based modeling and simulation approach. Hence, there have been some works in this direction.

Examples are MATSim [13] (which deals with activity-based simulation of route choice) and [14], [15] (based on reinforcement learning, Q-learning, and learning automata).

Route choice under various levels of information is turning a hot research topic due to the increasing use of navigation devices. Agent-based route choice simulation has been applied to investigate the effects of intelligent traveler information systems. The main questions here are what happens to the overall demand, if a certain share of drivers is informed and adapts. What kind of information is the best to give? Examples for such research line can be found in [16] for a two-route scenario, or in [17] where a neural net-based agent model for route choice is presented regarding a three route scenario.

One problem with these approaches is that their application in networks with more than a couple of routes between a few locations is not trivial. One first issue is that a set of reasonable route alternatives has to be generated. An k-shortest path algorithm can be used but it may output routes that differ only marginally. Additionally, all approaches, including agent-based ones, consider one route as one complete option to choose. On-the-fly re-routing has just started being considered.

To address this issue, re-routing in a scenario with multiple origins and destinations was studied in [18]. Besides route choice by the driver agents, the authors also consider traffic lights as adaptive agents in order to test whether such a form of co-adaptation may result in interferences or positive cumulative effects. This was one of the first works in the agent-based community that has dealt with agents computing new routes on the fly. This is important because en-route modifications cannot be ignored in a realistic simulation of decision making in traffic. An abstract route choice scenario was used, having some features of real world networks. However, in this work no comparison is made to methods that approximate the user equilibrium thus, it is not possible to fully assess the quality of those results.

Finally, we remark that there are implications of using one or the other class of methods (assignment versus route choice). For example, degradation in performance caused by selfish route choice can be measured by the so-called price of anarchy [19], [1], in which authors show results for small networks such as the one used to illustrate the Braess paradox [2].

IV. Approach and Case Study

One of the problems with the iterative methods discussed in Section II is that routes need to be recomputed at each iteration, to take into account the real cost of links. However, this information may be hard to get. Even state-of-the-art navigation devices do not have this information in a complete and accurate way. Also, in the successive averages method, the computational cost of the method depends on a good choice of the parameter ϕ. If it is a function of the parameter n (see Section II), then the efficiency may be compromised. Moreover, due to drivers' preferences, it is not clear that real-world drivers would follow the assumption of these methods, namely that all driver select the route with the least travel time.

The approach proposed here to compute the system optimum is GA-based and works in a different way. Given a number of vehicles, each having an origin and a destination (OD pair), and k shortest paths computed for each OD pair, an optimization process searches for one shortest path (among the k ones) for each vehicle so that the average travel time is minimized. This is done by trying new solutions that differ from those in the previous population of solutions, whereas a solution is a set of route choices, one for each vehicle. Diversity is achieved by mutation and crossover.

This is far from trivial because links have travel times that depend on the number of vehicles using them, thus potentially each choice affects the travel times of many other vehicles. As said, in order to do the optimization, our model considers that a chromosome represents the choice of route for each vehicle. Thus the length of the chromosome is the number of vehicles and each position can take one out of k different values.

Two remarks must be made here. First it is obvious that this method does not scale well with the number of vehicles, especially if k is large. Two, complementary, solutions for this are: (i) keep k low; in fact, k is typically no more than a couple of paths (either because there are indeed not many significantly different paths in real-world networks, or because drivers do not know them and/or their costs with accuracy); (ii) the number of vehicles to be put in the chromosome can be reduced by grouping them (e.g., a number of vehicles with have similar origins and destinations and/or similar driving patterns can be grouped so that a path will be assigned for the group).

The second remark regards the representation of the problem (chromosome); for this one could propose that instead of the just mentioned representation, a different one could be that each position in the chromosome would be a link and the values would be the number of vehicles using each link. However this hides the choices of each individual vehicle rendering the actual computation of the travel time impossible. Therefore the first mentioned representation is used here.

In order to illustrate the approach, a non-trivial network is used, namely the one suggested in Chapter 10 of [3] (Exercise 10.1) and depicted in Figure 1. Henceforth this network is referred as OW network.

In this network all links are two-way. The network represents two residential areas (nodes A and B) and two major shopping areas (nodes L and M). The numbers in the links are their travel times under free flow (the travel time a vehicle needs to travel the link *under no congestion*). For the shortest path algorithms, these can be seen as their costs thus henceforth both terms are used.

For the aforementioned scenario, authors in [3] use 1700 cars, i.e., this is the proposed demand for the OW network during a Saturday morning peak and an estimated demand from A and B to L and M as depicted in the second column of Table I.

If there would be no congestion, each vehicle would travel at its free flow travel time. These times are shown in the fourth column of Table I, i.e., they are computed when $k = 1$. The other columns show travel times for up to $k = 4$, as well as the corresponding paths.

However, the assumption of no congestion is unrealistic in this scenario because 1700 vehicles cannot use the links simultaneously without causing congestion. For this case, the book proposes a VDF that relates cost $c(q_k)$ at link k to its flow q_k. Specifically, it is proposed that the travel time in each link is increased by 0.02 for each vehicle/hour of flow ($t_k = t_{k_0} + 0.02 \cdot q_k$, as discussed in Section II). That is why we use this simple VDF, and remind the reader that more sophisticated ones (e.g., exponential relationship between flow and delay) could easily be used as well.

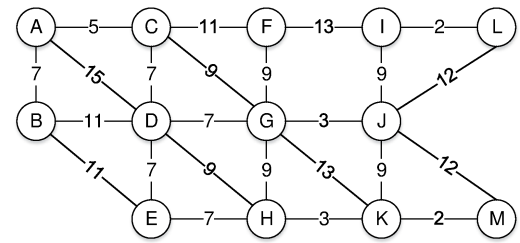

Fig. 1. Original Road Network (as Ortuzar and Willumsen).

To give an idea of travel times in this case, if all vehicles act rationally and use their shortest paths, the average travel time for all vehicles would increase to 89. This however is just part of the picture because, for instance, vehicles in the AL flow are hit harder: instead of the expected travel time of 28, the average travel time for AL vehicles would be 114. This explains why a good assignment is so important in this scenario.

This simple scenario goes far beyond simple two-route (binary) choice scenarios that are common in the literature. It captures properties of real-world scenarios, like interdependence of routes with shared links, heterogeneous demand throughout the complete network, and it has more than a single OD pair. Hence, it is hardly possible to compute the equilibrium algebraically.

Therefore we employ a GA to find an approximate solution, as the iterative methods also do (but then, for the user equilibrium). Given a population containing p chromosomes, each containing a candidate solution where each vehicle is assigned a path, the GA evolves the population so that the average travel time (the fitness function) is minimized. The evolution is made in a standard way with selection for crossover pairing with probability c and a mutation probability p_m. The software pyevolve (http://pyevolve.sourceforge.net/) was used.

V. Experiments and Results

Experiments were conducted using both the iterative methods discussed in Section II, and the GA-based approach proposed here. Results for incremental and successive averages methods stem from our implementation of these algorithms.

Results for the iterative methods discussed before appear in Table II, which summarizes the average travel times per OD pair for both methods, as well as the average travel times over all trips. These values were obtained using $p_n = 0.4, 0.3, 0.2, 0.1$ for the incremental method, and $\phi = 1/n$ and 100 iterations as stop criterion. It is important to remark that, at each iteration, new shortest paths are computed because

TABLE I
$k = 4$ SHORTEST PATHS AND THEIR FREE-FLOW TRAVEL TIMES (FFTT) FOR THE FOUR OD PAIRS.

OD	Trips	sh. path 1		sh. path 2		sh. path 3		sh. path 4	
AL	600	ACGJIL	28	ACGJL	29	ACFIL	31	ACDGJL	34
AM	400	ACDHKM	26	ACGJKM	28	ACGJM	29	ADHKM	29
BL	300	BDGJIL	32	BDGJL	33	BACGJIL	35	BACGJL	36
BM	400	BEHKM	23	BDHKM	25	BDEHKM	30	BDGKM	33

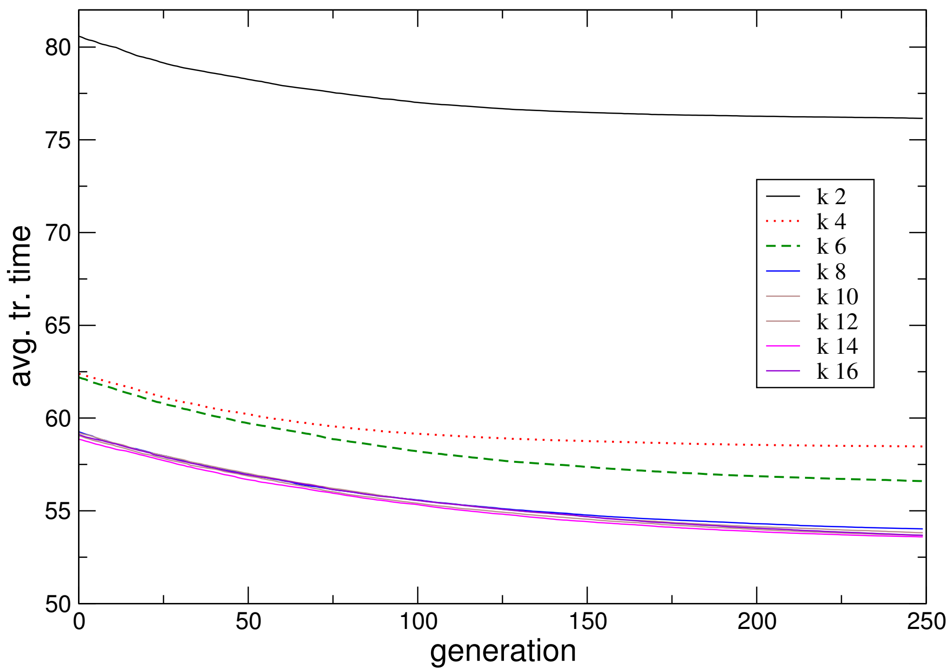

Fig. 2. Performance along generations for various values of k ($p_m = 0.001$ and $c = 0.2$).

TABLE II
AVERAGE TRAVEL TIME (OVERALL AND PER OD PAIR) FOR THE ITERATIVE METHODS (USER EQUILIBRIUM)

OD	Incremental	Succ. Avgs.
AL	69.92	70.92
AM	58.36	64.82
BL	73.24	68.90
BM	57.36	62.22
all	64.83	67.08

after each iteration, the flow distribution changes due to the route choices in the previous iteration step. Depending on the discretization of p_n and of how ϕ is set, the computational effort can be high.

In the case of the GA-based approach, recomputation of the shortest paths is not done. The k shortest paths are computed once and remain unchanged. These k shortest paths under free-flow (Table I), which are necessary to run the GA, were computed by the algorithm proposed by Yen [20].

Before we discuss the results, we remark that for the GA-based approach, the standard deviation is very low (close to zero once the convergence was established). Running times are in the order of few seconds for the iterative methods (this however can be higher if other values for the parameters were set). In the case of the GA-based approach, running times depend on the number of generations. For the cases shown next, simulations take at most a few minutes. Experiments were run in a standard PC (8 GB RAM), running Linux.

For assessment of the proposed method, here we discuss three possibilities. First, the performance measure is the same as used in [3]: travel times averaged over all vehicles and also over vehicles in each OD pair. Since the average travel

TABLE III
NUMBER OF TRIPS OVER SELECTED LINKS, FOR FOR $c = 0.2$, $p_m = 0.001$, $k = 8$

Link	Incr.	Succ. Avg.s	GA	Link	Incr.	Succ. Avg.s	GA
BA	60	3	234	DG	500	379	442
CD	160	4	244	FI	240	400	347
JM	40	128	93	GK	160	80	18

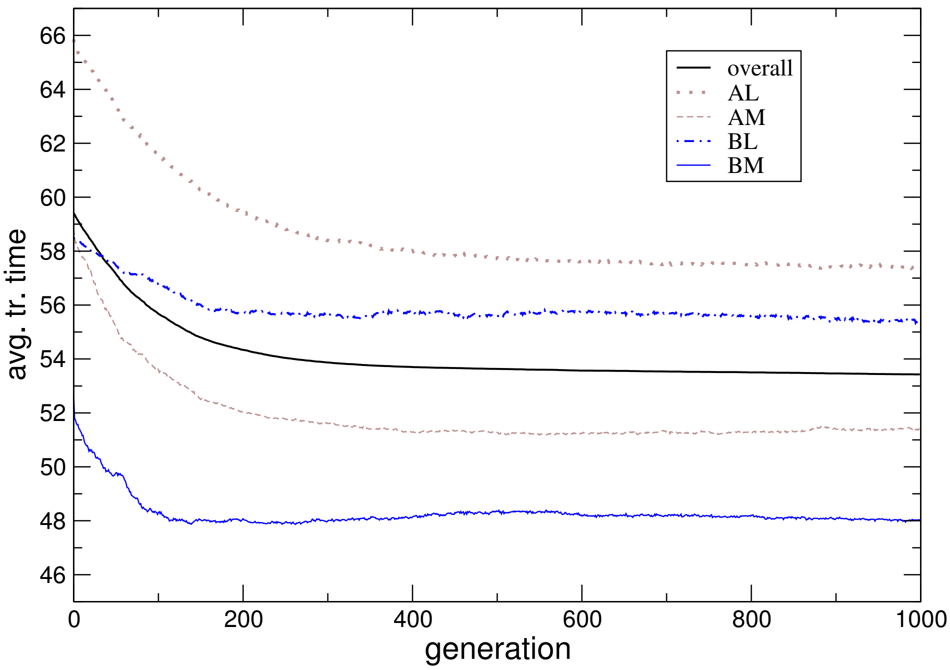

Fig. 3. Performance along generations: average travel time over all trips, as well as per OD flow (here $k = 8$).

time cannot say much about the behavior of the worst travel times, a second set of simulations were performed, in which the objective is to minimize the worst travel time (among all 1700 cars). However, since the worst travel time is likely to be always associated with the same flow (in this case, AL, as discussed next), in a further simulation set, the objective was set to minimize the sum of the four worst travel times, one in each flow. All these possibilities are discussed after we present the values for the parameters of our model.

Regarding the parameters of the GA-based approach, selection was done using standard methods, with crossover rate of 0.2 and mutation rate of 0.001. Note that these values were selected after extensive tests with other values. To give some hints about results achieved with other values, we show such results in Table IV. After these tests, it is possible to state that higher mutation rates are deleterious, while the crossover rate has low influence. It seems that if mutation is higher than the one we are using, choices are too random and thus convergence is harder.

In Figure 2 we show plots of the fitness value over gener-

TABLE IV
AVERAGE TRAVEL TIME FOR DIFFERENT VALUES OF MUTATION AND CROSSOVER AFTER 1000 GENERATIONS ($k = 8$)

mutation p_m	crossover c			
	0.05	0.1	0.2	0.3
0.001	53.50	53.44	53.34	53.50
0.01	54.11	54.22	54.01	54.08
0.1	56.53	56.40	56.49	56.44

TABLE V
AVERAGE TRAVEL TIME (OVERALL AND PER OD PAIR) FOR THE GA-BASED APPROACH (PARAMETERS FOR THE GA: $k = 8$, $p_m = 0.001$, $c = 0.2$)

OD	GA
AL	57.37
AM	51.44
BL	55.35
BM	48.05
all	53.43

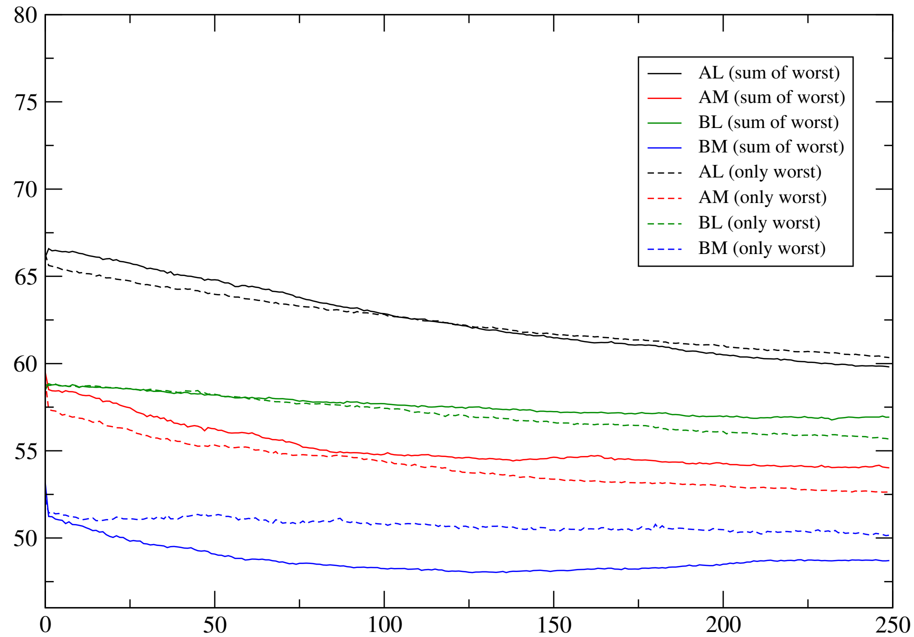

Fig. 4. Performance along generations: average travel time per OD flow (here $k = 8$) when the objective function is to minimize worst travel time (dashed lines) or to minimize the four worst travel times, one for each flow (dotted lines).

ation for various values of k, when the objective function is to minimize the average travel time. Note that depending on the value of k, already a random choice of routes for each vehicle can achieve results that are better than those achieved running the iterative methods. For instance, if a random choice of paths is performed for $k = 4$, the average travel time over all vehicles is 62. This is not quite the case if $k = 2$ (each vehicle can select between only two routes), possibly because a random choice with only two options is not beneficial.

In Figure 2 it is possible to see that the value of k affects the performance of the GA. If k is low then cars have few options of routes. Not only this: from Table I one can see that less links are involved in the shortest paths when $k = 2$ (as compared, e.g., with the case when $k = 4$).

The consequence is clear: less diversity in options for combinations of routes leads to many cars using the same links. This leads to higher travel times for everyone.

Figure 2 shows also that there is no significant difference in performance or convergence pattern of the GA if $k \geq 8$. Therefore, for this problem, $k = 8$ is used to run the GA-based approach for 1000 generations in order to see what happens after the timestep 250.

Results are depicted in Table V (average travel time after 1000 generations, both for all cars and for cars in each OD pair), and in Figure 3 (plots along generations). One sees the plot of the average travel time over all vehicles, as well as for each of the four flows (OD pairs). This is important to show that the average travel time was improved (over generations) for all flows, i.e., there was no improvement for given flow(s) at expenses of other(s).

It is possible to notice that the decrease in travel time is smooth for flows starting at node A, whereas the convergence for flows starting at B fluctuate for a while. The reason is that link BA appears in several shortest paths of flows starting at B, thus vehicles departing from B and using BA have to compete with vehicles departing from A. Conversely, link AB appears in few shortest paths for flows starting at node A.

When the objective function is set to minimize the worst travel time (which, we remark, is likely to be of a car in flow AL as shown in Figure 3) or the sum of the worst 4 travel times (one for each flow), one sees slightly different patterns, depending on the value of k. Here we discuss the case of $k = 8$.

Figure 4 compares results obtained for those two objective functions, for $k = 8$. Specifically, we plot the average travel time for each of the four flows. Dashed lines refer to the first case (minimize the worst travel time), while solid lines refer to the sum of the 4 worst travel times.

In flow AL it makes no significant difference to minimize the worst travel time or the sum of the worst 4 travel times. For BM it is better to minimize the sum of four worst travel times, i.e, using this objective function, the average travel time of cars in flow BM is lowest when we minimize the sum of worst travel times in each flow (not only the travel time of the worst car). However, for flows AM and BL, it is the opposite.

We must also say that there is not a big difference in the average travel times (over all 1700 cars) when we compare these travel times at generation 250, in case the objective function is: a) minimize the average travel time (Figure 3); b) minimize the worst travel time (Figure 4, dashed lines); and c) minimize the sum of the worst four travel times (Figure 4, solid lines). Rather, a pattern that does emerge is the fact that each of these objective functions has implications at some flows, i.e., one objective function may lead to a increase in travel time in one flow but decrease in travel time in another flow. This is expected because these flows share links and each objective function will just distribute cars over links in a different way but they do not eliminate the sharing of links. This way, an extension of this work is the study of multi-objective optimization.

Apart from path travel times, it is interesting to check what happens in each link. We remark that the number of trips using some links differs significantly in the incremental and in the successive averages methods. Thus, a direct comparison with the GA-based approach is interesting. Table III shows the number of trips in selected links (the criterion for inclusion in this table was that the link travel time differs in both iterative methods). For instance, for the link BA, the incremental and successive average methods assign 60 and 3 trips respectively. The GA-based approach assigns 234 trips.

We note that these results refer to $k = 8$. Other values of k result in different sets of links appearing in the shortest paths (the lower the k, the lower the diversity). Therefore values in Table III can be different for different values of k.

VI. Conclusions and Future Work

Traffic assignment is an important step in modeling transportation systems. In this paper we use GA to compute an approximate solution to the trip assignment, which minimizes the average travel time. Contrarily to other works (especially those in the multiagent community), we report results for a non-trivial network. The example used here is rich in the sense that it does not deal with a single OD pair and a trivial binary route choice.

Results have shown that the GA-based optimization is able to compute fast the assignment that minimizes the average travel time for all cars, as well as for the worst cars in each flow.

A future direction of this work is to combine the presented method with local search and perform multi-objective optimization.

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